

DINUJA PERERA

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 2+ years of experience designing, building, and deploying supervised, unsupervised, and generative AI models. Hands-on experience with Azure Databricks and GPU-accelerated computing. Currently working as a KTP Associate at Uniper UK in collaboration with Loughborough University, developing AI solutions for gas power plant systems. Holder of a Tier 1 Global Talent visa, endorsed by UK Research and Innovation (UKRI), with full UK work rights and open to relocation.

WORK EXPERIENCE

KTP Associate – Machine Learning Engineer (Predictive Maintenance) | Uniper UK | Loughborough University - June 2024 - present

- Built and tested generative AI models (LSTM, ESN, RNN) on turbine sensor time-series data to forecast signals and detect early signs of gas turbine trips.
- Applied unsupervised learning (K-Means clustering) to group sensor behaviours and improve feature engineering for anomaly detection.
- Implemented a statistical correlation-based model to identify faulty instruments, comparing autocorrelation patterns of healthy vs. faulty sensors.
- Ran proof-of-concept experiments with real turbine data to check whether models were applicable to operational use cases before scaling further.
- Processed and analysed large-scale sensor datasets in Azure Databricks using GPU acceleration to speed up deep learning training.
- Designed interactive dashboards to share diagnostic results and visual insights with engineers and managers.
- Worked closely with government, academic, and industry partners (data scientists, reliability engineers, plant operators) to align technical work with operational needs.

Machine Learning Engineer | CML Insight Inc., Texas, USA | Nov 2021 – Jan 2023

- Cleaned, merged, and standardised multiple datasets to ensure smooth integration for downstream analysis.
- Engineered NLP features (tokenisation, embeddings, sentiment scores) from text data to support model development.
- Assessed feature importance and checked for potential target leakage to improve model reliability.
- Developed cohort-based models to analyse behavioural patterns across different customer groups.
- Supported the team by onboarding and mentoring interns on basic ML workflows and coding practices.
- Practiced object-oriented programming in Python to improve code organisation and reusability.
- Maintained ML pipelines to enable model reuse on future incoming data.

Teaching Assistant | Department of Information Systems Engineering, University of Colombo, Sri Lanka | Jun 2019 - Nov 2021

- Conducted practical sessions and supervised projects for master's and undergraduate courses, including 'Machine Learning and Neural Computing, Data Analytics, and Embedded Systems.
- Co-ordinated grading assignments, conducting code reviews, and invigilating exams for undergraduate and master's level programs.

Intern Data Analyst | Brandix Apparel Limited | Jun 2019

- Analysed a large spare parts dataset, using Microsoft Excel Pivot Tables to detect data discrepancies and inaccuracies.
- Collaborated with cross-functional teams to address data quality issues and implement corrective actions, enhancing inventory management.

Technical Skills:

- **Languages and Frameworks:** Python, TensorFlow, Keras, PyTorch, Scikit-learn, PySpark, Pandas
- **Cloud :** Azure Databricks, Google Colab, AWS (basic)
- **Databases:** SQL, MongoDB, Cassandra, Redis, Neo4j
- **Analytics and Visualisation:** Power BI, Tableau, Matplotlib, Seaborn, Excel pivot

- **ML Models:** Generative AI, LSTM, ESN, NLP, RNN, Isolation Forest, Autoencoders, Regression Analysis, Random Forest, Decision Trees, SVM, KNN, Naïve Bayes, K-Means, PCA
- **DevOps and Collaboration Tools:** GitHub, ClickUp, Confluence, Azure Project Management

Research Projects:

- **Leveraging NLP for Enhanced Company Performance: Transformer-Based Sentiment Analysis and Comparative Review** - Implemented transformer-based sentiment analysis alongside topic modeling and aspect-based analysis, to provide actionable insights for companies to improve products, services, and organisational aspects.
- **Benchmark NLP Algorithm for Hate Speech Detection on Social Media** - Tested 12 deep learning architectures, including RNNs, CNNs, transformer-based models (e.g., BERT, RoBERTa), and hybrid architectures (e.g., CNN + LSTM) to detect hatespeech on social media platform.
- **Comparative NLP model analysis on Hate Speech Detection on Social Media Platforms** - Demonstrated RoBERTa's superior performance over BART in both binary and multi-class classification tasks. Highlighted its potential for high-resource languages and future research in low-resource languages and multi-modal data.

EDUCATION

MSc Data Science - Distinction | University of Greenwich, London | 2023 – 2024

- Key Modules: Machine Learning, Applied Machine Learning, Data Visualisation, Statistical Methods for Time Series Analysis, Graph and Modern Databases, Big Data, Blockchain for FinTech Applications.
- Completed projects involving comparative model evaluation (regression, classification, neural networks), clustering analysis, and optimisation of models for diverse datasets.

MSc specialising in Data Science, Analytics and Engineering | University of Moratuwa, Sri Lanka | 2021

- Key Modules: Machine Learning, Pattern Recognition, Data Mining, Data Analytics, Advanced Databases, Business Intelligence, Neural Networks, Advanced Algorithms, Statistical Inference, Bioinformatics.

Grad. Dip. in Electronics Telecommunication & Computer Engineering - Distinction | Institution of Engineers, Sri Lanka | 2020

- Attained a GPA of 3.75 (out of 4.20) in key modules including Digital Signal Processing, Computer Security, Computer Networks, and Communication Engineering. Implementing an IoT-based Air Quality Monitoring System for a remote Greenhouse using GSM modules, microcontrollers, and sensors.

BSc (Electronics & IT Theme) - Second Class Honours | University of Colombo, Sri Lanka | 2015-2019

- Earned a GPA of 3.24 (out of 4.00) in key modules including Applied Mathematics, Statistics, Computer Science, and Physics.

PUBLICATIONS / CONFERENCE PRESENTATION

"Short Comparative Analysis on Pretrained BART and RoBERTa in Detecting Hate Speech on YouTube and Reddit Platforms," presented at WiNLP Workshop co-located with EMNLP 2022.

EXTRACURRICULAR & CERTIFICATIONS

- Leadership & Management - CMI Level 5 Equivalent - Ashorne Hill Management College, UK
- British Airways Data Science Job Simulation on Forage - April 2024
- LinkedIn Learning Certifications: Advanced NLP, PySpark for Big Data, Azure DevOps Fundamentals, Generative AI Foundations

REFERENCE

- Mike Allen-Operations and Maintenance Manager, Enfield, Grain and Taylors Lane Power Stations, Uniper UK